Improving Motion Cueing with High Shelf Filters

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Abstract - *In audio engineering the "high shelf" can provide a sense of "clarity" or "air" to an audio signal. By increasing high-frequency gain, it is hypothesised that such clarity could also be brought to motion. To test this hypothesis, a second order high shelf filter is combined with the classical Motion Cueing Algorithm (MCA) on a state-of-the-art, high bandwidth, ultra-low latency Dynisma DMG-1 motion simulator. Subjective feedback is gathered from six highly-skilled drivers. Preliminary results hint towards the high shelf's ability to increase perceived power of turn-in and entry braking cues, fidelity of mid-exit balance cues as well as road surface detail. These improvements are attainable with reduced workspace usage than a 'full-spectrum' increase in gain or scaling of the same magnitude. Therefore, the technique is especially relevant to 'compact footprint', 'high bandwidth' motion simulators.*

Keywords: *ride, handling, drivability, ultra-low latency, high-bandwidth*

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Introduction

In audio engineering the "high shelf" can provide a sense of "clarity" or "air" to an audio signal (Winer, 2017). It is hypothesised that such clarity could also be brought to motion.

A generalised low shelf filter of order M is given in Holters and Zölzer (2006). To derive the equivalent high shelving filters, the following steps are required. Firstly, application of the conformal mapping $s \leftarrow s^{-1}$ which interchanges high and low frequencies (Holters and Zölzer, 2006; Smith, 2007). Finally, denormalisation by substituting s= sω^{-1}.

These substitutions produce Equation 1, which can be used to construct high shelf filters of positive integer order M, where g is the gain and ω is the transition frequency. By construction, these filters have minimum-phase behaviour, that is to say, all poles and zeros are located on the left side of the complex plane. This is important for motion simulator applications since phase and group delay are related to latency, which is key for (performance) driving (Melzi and Previati, 2025).



Figure 1: The "Air Band" is a high shelving filter known for its pleasing sonic qualities. With permission of Maag Audio.

$$H(s) = \prod_{m=1}^{M} \frac{\omega + \sqrt[M]{g}\left(e^{j\alpha_m}\right) s}{\omega + \left(e^{j\alpha_m}\right) s} \tag{1}$$

By setting $g = 1 + BK^{-1}$ simulator operators may conveniently treat the shelf as an offset or boost B to the overall vehicle acceleration scaling K.

Importantly, when placed in series with a classical washout filter, washout occurs as expected.

Magnitude and phase response for first and second order (dashed) high shelving filters, with $\omega = 10 \times 2\pi, g = 1.05, 1.15, 1.25$ are shown in Figure 2.

In summary, the shelf filter offers the ability to *scale high frequency content separately to overall motion*.

Methodology & Results

Possible applications of high shelving includes entry braking, corresponding to the longitudinal axis, handling, corresponding to yaw and lateral axes and ride, corresponding to vertical, roll and pitch axes.

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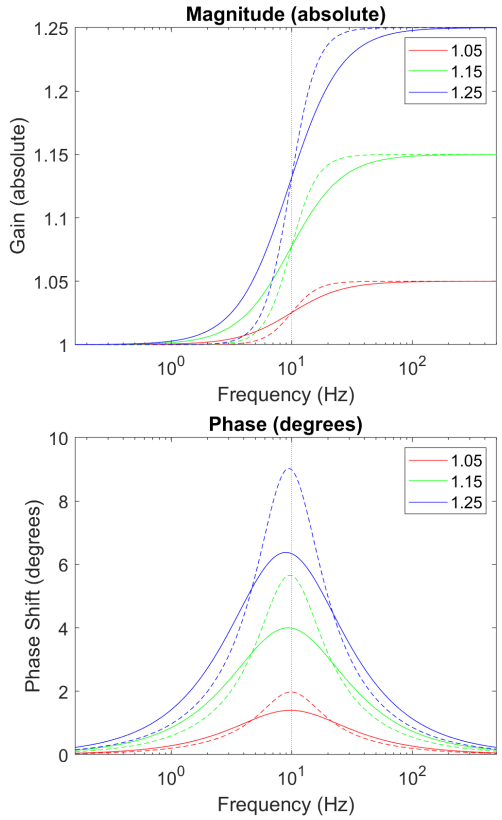


Figure 2: Comparison of high shelf filters for different values of g. Solid lines: 1st order. Dashed: 2nd order.



Figure 3: Dynisma DMG-1 test at Spa-Francorchamps.

An LMP2 vehicle model was used in AVL VSM. The model is well-correlated to a contemporary, competitive car. All drivers are highly skilled: three of the six were professional and hold motorsport licences.

Participants were instructed to drive full-pace, consistent laps of the Circuit of Spa-Francorchamps, and were blind to parameter changes.

Feedback was sought via an intercom system, asking the driver to rate changes on a numeric scale. Though informal, this is how technologies are typically assessed in a motorsport setting due to time pressures involved.

Results are summarised in Table 1 and are a collated summary of all feedback received.

Table 1: Table summarising driver feedback.

Axes, ω, B	**Comments (collated)**
Lateral 2Hz, Yaw 1Hz $B \leq 0.3$	Perceived increase in: 1) bite at corner entry, increased sense of rear stability and turn in power; 2) understeer mid corner—likely induced by contrast with entry feeling; 3) oversteer on corner exit with kerb strike.
Longitudinal [1Hz 2Hz], $B \leq 0.3$	Perceived increase in: 1) initial braking efficiency; 2) pitching (despite pitch filter being unchanged); 3) rear traction when applying throttle after braking. Heavy braking zones perceived as more realistic. Other braking zones perceived as less realistic (likely by contrast).
Vertical, Roll, Pitch, 2.5Hz, $B = 0.1$	Perceived increase in road surface detail and realism. Improved sense of connection to road.

Table 2: Dynisma DMG-1 specifications

Specification	**Value**
Nominal payload	250kg – 400kg
Motion bandwidth	Up to 100Hz
Motion latency	<5ms
Max. excursion (long, lat, heave)	1300mm, 1300mm, 150mm
Max. excursion (roll, pitch, yaw)	$\pm 8°$, $\pm 7.5°$, $\pm 45°$
Combined lat-yaw	500mm @ $\pm 30°$
Combined lat-long excursion rad.	650mm

Discussion & Conclusion

A new method of increasing realism in motion simulators was presented. It is simple, easy to implement, intuitive and may be placed in series with any MCA. It is especially powerful for high bandwidth simulators, such as the Dynisma DMG-1 used in this study (Table 2), as it increases motion fidelity without a significant increase in workspace usage. Whilst feedback gave a strong indicator of improvement, more testing is required to be conclusive.

The range of potential applications include, but are not limited to, improving perception of corner entry, rear stability, oversteer, understeer, braking efficiency, rear traction, road surface detail and Noise, Vibration & Harshness (NVH).

A combined classical/high shelving MCA has 6+ parameters per axis giving 36+ in total, which becomes difficult to tune manually. Automatic parametrisation (Asadi, et al., 2015), using a tool such as ProCue (Brown, 2024), is recommended for extracting the most performance from this new approach.

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